

Polymarket Gamma Starter Operator Manual

Version 1.0.1-real

Generated 2026-06-14

Local-first research, paper operations, staged live-readiness, and guarded manual execution control.

Classification	Value
Document purpose	Deep operator guide for installation, configuration, daily operation, live-readiness, guarded manual live submit/cancel, audit, maintenance, and troubleshooting.
Primary operator	Human operator with authority over configuration, credentials, risk limits, and final execution decisions.
Safety baseline	Fail-closed. Live trading, autonomous trading, real network, submit, cancel, scheduler, and real-live tests are disabled by default.
Financial advice	This software and manual are not financial advice. Operators are responsible for every real-money outcome.

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1. Critical Safety Notice

Read this section before starting the app, connecting it to a LAN, entering credentials, or enabling any live-adjacent setting.

- This software is not financial advice. It is an operator console and workflow application.
- Prediction markets are risky. A correct workflow does not guarantee a profitable or correct trade.
- Live trading is disabled by default. Autonomous live trading is not enabled in this release.
- The app contains guarded manual live submit/cancel code paths, but they require explicit local configuration, credentials, allowlists, limits, kill switch off, source records, and final confirmation.
- Automated validation and normal status screens must not submit or cancel real orders.
- Fake adapter receipts are local simulations only. They are not exchange acknowledgements.
- Market data and execution-quality outputs are estimates and can be stale, incomplete, or misleading.
- Keep private keys, API secrets, L2 credentials, sessions, user databases, generated ledgers, and state out of shared ZIPs and source-controlled folders.
- When uncertain, leave the kill switch active and use paper/fake-adapter workflows only.

Unsafe action	Required operator response
A page says ready but kill switch is active	Treat the action as blocked. Do not assume a hidden order was sent.
A fake-local receipt appears	Treat it as local validation only, not a Polymarket acknowledgement.
A live status page shows missing credentials	Do not attempt live submit/cancel. Add credentials only in local .env after reviewing current Polymarket docs.
Autonomous mode says live blocked	Expected. Autonomous live trading is not enabled in v1.0.1-real.
A real order status is unknown	Use live order ledger plus reconciliation. Do not duplicate orders without manual external verification.
Internal server error appears	Stop, capture URL/log, and use troubleshooting steps. Do not continue execution workflows until fixed.

2. System Overview and Operating Model

Polymarket Gamma Starter is a local-first, human-in-the-loop application. Its main job is to make market research, paper workflow, risk review, operator approval, execution readiness, and audit trail visible before any real execution decision is possible.

2.1 Core principles

Local-first state: most operator records are written under the local data directory.

Human-in-the-loop: paper tickets, approvals, preflight, live intents, authorizations, packets, dry-run receipts, adapter requests, and manual execution attempts are explicit records.

Separation of stages: research, paper simulation, live-readiness, fake-adapter validation, and real manual adapter calls remain separate.

Auditability: important state transitions can be listed, exported to CSV, and cross-checked through API/CLI/UI surfaces.

Fail-closed execution: live and autonomous paths block unless every gate is configured deliberately.

2.2 Capability posture

Capability	1.0.1-real posture
Research and evidence	Enabled as local operator workflow.

Capability	1.0.1-real posture
Paper trading	Enabled as simulation/local ledger workflow.
Market data intelligence	Local/manual and optional public-fetch boundary; public fetch disabled by default.
Execution quality simulator	Deterministic estimates; not fill guarantees.
Live config/readiness	Enabled as local gated review.
Manual live submit/cancel	Implemented in code behind the CLOB adapter boundary, but disabled by default and not exercised by normal tests.
Autonomous live trading	Not enabled; remains dry-run/fake-adapter oriented and blocked by default.
Scheduler	Disabled by default; no uncontrolled background trading loop.

3. Installation and First Run

3.1 Recommended directory layout

Install into a dedicated project folder. Keep the release ZIP, extracted runtime folder, and backups separate. Never run directly out of a shared downloads folder that also contains credentials or generated state.

```
mkdir -p ~/polymarket-gamma
cd ~/polymarket-gamma
unzip polymarket-gamma-starter-v1.0.1-real.zip
cd polymarket-gamma-starter-v1.0.1-real
```

3.2 Python environment

Use a virtual environment for dependencies. The virtual environment must not be packaged into release ZIPs or uploaded as project source.

```
python -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
# Optional live-runtime only:
pip install -r requirements-live-optional.txt
```

3.3 Environment file and start command

```
cp .env.example .env
# edit .env locally; never paste private keys into chat or issue trackers
python run.py
# or
uvicorn app.main:app --host 0.0.0.0 --port 8000 --reload
```

23. Create the initial admin user at /setup. Use a strong unique password.
24. Log in as admin.
25. Open /administration/config and verify runtime warnings.
26. Confirm /health returns ok and version 1.0.1-real.
27. Confirm live status pages show safe defaults: live disabled, real network disabled, submit/cancel disabled, kill switch active.
28. Create or import research/watchlist data before expecting populated workflow pages.

4. Authentication, Roles, and LAN Deployment

The package includes local authentication. User records are stored in data/users.json. Session secret material is stored in data/session_secret.txt. Both are runtime state and must be excluded from release packages and backups that are not encrypted.

Role	Intended use
admin	Can manage users and perform administrative actions. At least one active admin must remain.
read_only	Intended for review/visibility workflows. The application still relies on route-level behavior and operator discipline.

4.1 LAN and security settings

Setting	Default	Operator guidance
HOST	0.0.0.0	Allows LAN devices to reach the app. Use 127.0.0.1 for local-only operation.
PORT	8000	HTTP port.
ALLOWED_HOSTS	*	Convenient for LAN testing; restrict to fixed hostnames/IPs for stable deployment.
SECURITY_HEADERS_ENABLED	true	Keep enabled unless debugging.
SESSION_COOKIE_SECURE	false	Normal for http:// LAN testing; set true behind HTTPS.

- Use a dedicated runtime user.
- Restrict firewall access.
- Require authentication before LAN exposure.
- Use HTTPS if leaving localhost.
- Keep .env permissions restrictive.
- Back up data/ securely.

5. Configuration Deep Dive

The app is configured primarily by .env. Many live-related fields are intentionally duplicated for backward compatibility across the staged release history. Use the newer POLYMARKET_LIVE_* names and leave legacy LIVE_* values safe unless a module requires them.

Key	Default/example	Purpose
HOST	0.0.0.0	Interface to bind. Use 127.0.0.1 for local-only or 0.0.0.0 for LAN.
PORT	8000	HTTP port.
APP_RELOAD	true	Development reload behavior.
ALLOWED_HOSTS	*	Allowed host header list.
APP_MODE	read_only	Display/runtime mode label.
READ_ONLY	true	Global read-only posture for core app surfaces.
LIVE_TRADING_ENABLED	false	Legacy broad live flag. Keep false unless deliberately operating live.
GAMMA_BASE_URL	https://gamma-api.polymarket.com	Polymarket Gamma API base.
CLOB_BASE_URL	https://clob.polymarket.com	CLOB API base used by non-live client/read-only tools.
REQUEST_TIMEOUT_SECONDS	20	HTTP timeout.
DEFAULT_LIMIT	20	Default list limit.

5.1 Paper risk settings

Key	Default/example	Purpose
PAPER_MAX_STAKE_PER_TRADE	250	Paper simulation risk guard. Applies to local paper workflow only.
PAPER_MAX_MARKET_EXPOSURE	500	Paper simulation risk guard. Applies to local paper workflow only.
PAPER_MAX_TOTAL_EXPOSURE	2500	Paper simulation risk guard. Applies to local paper workflow only.
PAPER_MAX_OPEN_POSITIONS	20	Paper simulation risk guard. Applies to local paper workflow only.
PAPER_MIN_LIQUIDITY	1000	Paper simulation risk guard. Applies to local paper workflow only.
PAPER_MIN_VOLUME_24HR	10	Paper simulation risk guard. Applies to local paper workflow only.
PAPER_BLOCK_EXTREME_PRICES	true	Paper simulation risk guard. Applies to local paper workflow only.
PAPER_MIN_PRICE	0.02	Paper simulation risk guard. Applies to local paper workflow only.
PAPER_MAX_PRICE	0.98	Paper simulation risk guard. Applies to local paper workflow only.

5.2 Live execution hard gates

Key	Default/example	Operational meaning
POLYMARKET_LIVE_MODE	false	Hard gate for live submit/cancel or live smoke tests. Leave default unless deliberately operating a controlled live window.
POLYMARKET_LIVE_ALLOW_REAL_NET WORK	false	Hard gate for live submit/cancel or live smoke tests. Leave default unless deliberately operating a controlled live window.
POLYMARKET_LIVE_ENABLE_SUBMIT	false	Hard gate for live submit/cancel or live smoke tests. Leave default unless deliberately operating a controlled live window.
POLYMARKET_LIVE_ENABLE_CANCEL	false	Hard gate for live submit/cancel or live smoke tests. Leave default unless deliberately operating a controlled live window.
POLYMARKET_LIVE_REQUIRE_MANUAL_ AUTH	true	Hard gate for live submit/cancel or live smoke tests. Leave default unless deliberately operating a controlled live window.
POLYMARKET_LIVE_KILL_SWITCH	true	Hard gate for live submit/cancel or live smoke tests. Leave default unless deliberately operating a controlled live window.
POLYMARKET_LIVE_MAX_ORDER_NOTI ONAL	0	Hard gate for live submit/cancel or live smoke tests. Leave default unless deliberately operating a controlled live window.
POLYMARKET_LIVE_MAX_DAILY_NOTIO NAL	0	Hard gate for live submit/cancel or live smoke tests. Leave default unless deliberately operating a controlled live window.
POLYMARKET_LIVE_MAX_OPEN_ORDER S	0	Hard gate for live submit/cancel or live smoke tests. Leave default unless deliberately operating a controlled live window.

Key	Default/example	Operational meaning
POLYMARKET_LIVE_MARKET_ALLOWLIST		Hard gate for live submit/cancel or live smoke tests. Leave default unless deliberately operating a controlled live window.
POLYMARKET_LIVE_TOKEN_ALLOWLIST		Hard gate for live submit/cancel or live smoke tests. Leave default unless deliberately operating a controlled live window.
POLYMARKET_LIVE_FINAL_CONFIRMATION_PHRASE		Hard gate for live submit/cancel or live smoke tests. Leave default unless deliberately operating a controlled live window.
POLYMARKET_LIVE_FAKE_ADAPTER_ENABLED	false	Hard gate for live submit/cancel or live smoke tests. Leave default unless deliberately operating a controlled live window.
POLYMARKET_LIVE_EMERGENCY_CANCEL_ENABLED	false	Hard gate for live submit/cancel or live smoke tests. Leave default unless deliberately operating a controlled live window.
POLYMARKET_RUN_REAL_LIVE_TESTS	false	Hard gate for live submit/cancel or live smoke tests. Leave default unless deliberately operating a controlled live window.
POLYMARKET_REAL_LIVE_TEST_CONFIRMATION		Hard gate for live submit/cancel or live smoke tests. Leave default unless deliberately operating a controlled live window.

5.3 Credential fields

Field	Manual notes
POLYMARKET_CHAIN_ID	Sensitive if it contains key, secret, passphrase, private key, wallet identity, or funder identity. Status pages should only show presence/redacted values.
POLYMARKET_SIGNATURE_TYPE	Sensitive if it contains key, secret, passphrase, private key, wallet identity, or funder identity. Status pages should only show presence/redacted values.
POLYMARKET_FUNDER_ADDRESS	Sensitive if it contains key, secret, passphrase, private key, wallet identity, or funder identity. Status pages should only show presence/redacted values.
POLY_ADDRESS	Sensitive if it contains key, secret, passphrase, private key, wallet identity, or funder identity. Status pages should only show presence/redacted values.
POLYMARKET_WALLET_ADDRESS	Sensitive if it contains key, secret, passphrase, private key, wallet identity, or funder identity. Status pages should only show presence/redacted values.
POLY_PRIVATE_KEY	Sensitive if it contains key, secret, passphrase, private key, wallet identity, or funder identity. Status pages should only show presence/redacted values.
POLYMARKET_PRIVATE_KEY	Sensitive if it contains key, secret, passphrase, private key, wallet identity, or funder identity. Status pages should only show presence/redacted values.
POLY_API_KEY	Sensitive if it contains key, secret, passphrase, private key, wallet identity, or funder identity. Status pages should only show presence/redacted values.
POLYMARKET_CLOB_API_KEY	Sensitive if it contains key, secret, passphrase, private key, wallet identity, or funder identity. Status pages should only show presence/redacted values.
CLOB_API_KEY	Sensitive if it contains key, secret, passphrase, private key,

Field	Manual notes
	wallet identity, or funder identity. Status pages should only show presence/redacted values.
POLY_SECRET	Sensitive if it contains key, secret, passphrase, private key, wallet identity, or funder identity. Status pages should only show presence/redacted values.
POLYMARKET_CLOB_SECRET	Sensitive if it contains key, secret, passphrase, private key, wallet identity, or funder identity. Status pages should only show presence/redacted values.
CLOB_SECRET	Sensitive if it contains key, secret, passphrase, private key, wallet identity, or funder identity. Status pages should only show presence/redacted values.
POLY_PASSPHRASE	Sensitive if it contains key, secret, passphrase, private key, wallet identity, or funder identity. Status pages should only show presence/redacted values.
POLYMARKET_CLOB_PASSPHRASE	Sensitive if it contains key, secret, passphrase, private key, wallet identity, or funder identity. Status pages should only show presence/redacted values.
CLOB_PASSPHRASE	Sensitive if it contains key, secret, passphrase, private key, wallet identity, or funder identity. Status pages should only show presence/redacted values.

5.4 Autonomous settings

Key	Default/example	Guidance
POLYMARKET_LIVE_ENABLE_AUTONOMOUS	false	Leave default/off unless running deliberate dry-run/fake-adapter signal tests. Autonomous live is not enabled in this release.
POLYMARKET_AUTONOMOUS_MAX_ORDERS_PER_RUN	0	Leave default/off unless running deliberate dry-run/fake-adapter signal tests. Autonomous live is not enabled in this release.
POLYMARKET_AUTONOMOUS_MAX_ORDERS_PER_DAY	0	Leave default/off unless running deliberate dry-run/fake-adapter signal tests. Autonomous live is not enabled in this release.
POLYMARKET_AUTONOMOUS_MIN_SIGNAL_CONFIDENCE		Leave default/off unless running deliberate dry-run/fake-adapter signal tests. Autonomous live is not enabled in this release.
POLYMARKET_AUTONOMOUS_REQUIRE_MARKET_DATA	true	Leave default/off unless running deliberate dry-run/fake-adapter signal tests. Autonomous live is not enabled in this release.
POLYMARKET_AUTONOMOUS_REQUIRE_EXECUTION_QUALITY	true	Leave default/off unless running deliberate dry-run/fake-adapter signal tests. Autonomous live is not enabled in this release.
POLYMARKET_AUTONOMOUS_REQUIRE_PAPER_APPROVAL	true	Leave default/off unless running deliberate dry-run/fake-adapter signal tests. Autonomous live is not enabled in this release.
POLYMARKET_AUTONOMOUS_STRATEGY_ALLOWLIST		Leave default/off unless running deliberate dry-run/fake-adapter signal tests. Autonomous live is not enabled in this release.
POLYMARKET_AUTONOMOUS_DRY_RUN_BY_DEFAULT	true	Leave default/off unless running deliberate dry-run/fake-adapter signal tests. Autonomous live is not enabled in this release.

Key	Default/example	Guidance
POLYMARKET_AUTONOMOUS_SCHEDULER_ENABLED	false	Leave default/off unless running deliberate dry-run/fake-adapter signal tests. Autonomous live is not enabled in this release.
POLYMARKET_AUTONOMOUS_SCHEDULER_INTERVAL_SECONDS	0	Leave default/off unless running deliberate dry-run/fake-adapter signal tests. Autonomous live is not enabled in this release.
POLYMARKET_AUTONOMOUS_SCHEDULER_DRY_RUN_ONLY	true	Leave default/off unless running deliberate dry-run/fake-adapter signal tests. Autonomous live is not enabled in this release.

6. Operator Console Navigation

The console is organized around visible workflow stages. Use the browser for context and forms; use CLI/API for repeatable audit, exports, and smoke checks.

Page	Purpose
/	Dashboard / home
/operator	Operator brief
/workflow	Workflow map
/administration/config	Runtime config/status
/users	User management
/opportunities	Opportunity scan
/research/sources	Research sources
/research/evidence	Evidence
/playbooks	Playbooks
/trade-tickets	Trade tickets
/preflight	Paper preflight
/approvals	Approvals
/execution-queue	Execution queue
/positions	Paper positions
/settlements	Settlements
/risk-budget	Risk budget
/audit	Audit
/market-data	Market-data intelligence
/execution-quality	Execution-quality simulator
/live-config	Live config readiness
/live-order-intents	Live order intents
/live-order-intent-preflight	Live preflight
/live-order-authorizations	Live authorizations
/live-execution-packets	Execution packets
/live-dry-run-adapter	Dry-run adapter
/live-adapter-requests	Adapter requests
/live-clob-adapter	CLOB adapter
/live-trading	Live trading status
/live-orders	Live order ledger
/live-reconciliation	Live reconciliation
/strategy-signals	Strategy signals
/autonomous-trading	Autonomous trading bridge

- Badges such as LOCAL, PAPER ONLY, LIVE DISABLED, READ ONLY, FAKE ADAPTER, LIVE GUARDED, KILL SWITCH ACTIVE, and AUTONOMOUS OFF are safety-critical cues.
- A populated table means records exist, not that execution occurred.

- CSV links are part of the audit surface.
- If a page is empty on first run, create upstream records first.
- If a page returns an internal error, stop and investigate before continuing.

7. Research and Evidence Workflow

Research is the first decision layer. It should answer whether the market deserves paper workflow attention, whether the thesis is clear, and what evidence could invalidate the trade.

29. Open /opportunities or use --opportunity-engine to rank markets.
30. Review prices, liquidity, volume, time to expiry, and scoring/edge signals.
31. Add promising markets to watchlist rather than immediately creating tickets.
32. Create market notes for thesis, counter-thesis, and invalidation.
33. Use /research/sources and /research/evidence to collect and inspect evidence packets.
34. Use evidence scoring and evidence-adjusted probability as review aids, not automatic signals.

Research artifact	Purpose	Common next step
Watchlist row	Keep a market visible for repeated review.	Evidence packet or playbook fit.
Evidence packet	Bundle sources and observations.	Score packet and review probability.
Evidence score	Evaluate evidence quality.	Create ticket only when thesis justifies it.
Market note	Operator rationale and observations.	Use as ticket rationale.
Source status	Check source availability.	Avoid acting on missing/broken sources.

8. Paper Trading Lifecycle

Paper trading is a local simulation workflow. It validates process, thesis quality, risk discipline, and operational handoff before live-readiness is considered.

35. Research market and create evidence/notes.
36. Evaluate playbook fit and readiness.
37. Create a trade ticket.
38. Run preflight and risk budget review.
39. Approve or reject the ticket.
40. Execute paper buy only after approval and preflight allow it.
41. Monitor paper position.
42. Create exit ticket or update position plan.
43. Execute paper sell or settle when appropriate.
44. Review audit and closeout records.

Paper action	Recommended gate	Result
Create ticket	Research and playbook fit complete	Local ticket created.
Preflight ticket	Risk limits and market constraints reviewed	Ready/warning/blocked state.
Approve ticket	Human approval after preflight	Approval record.
Paper buy	Approved ticket and acceptable risk	Local paper trade and position updated.
Create exit ticket	Exit thesis or risk trigger	Local exit plan.
Paper sell	Exit ticket or manual decision	Local paper position reduced/closed.
Settle	Market outcome known	Local paper P/L finalized.

9. Risk, Preflight, Approvals, Execution Queue, and Positions

Paper risk budget summarizes exposure, limit usage, and market-specific risk. It is a rehearsal for live control, not a box-checking exercise.

State category	Operator meaning
ready	No blocking conditions detected; continue only with human review.
ready_with_warnings	Warnings exist; document decision.
blocked	Do not execute; review blockers and update upstream record or risk config.
invalid/missing	Required source record is absent or malformed.

- Approvals are human decision records and do not override stale preflight/risk gates.
- Execution queue lists approved/actionable/blocked/waiting tickets.
- The /positions page is the paper portfolio command center and is hardened in v1.0.1-real against malformed local paper data.
- Use position plans, alerts, and exit tickets for repeatable management.

10. Paper Operations, Handoffs, Escalations, and Closeout

The paper ops layer manages multiple markets, sessions, and unresolved decisions without losing context.

- Open /paper-ops-briefing at session start.
- Record checkpoints as session progresses.
- Use handoffs for unresolved work between sessions.
- Use aging boards to find stale decisions.
- Use escalations for blocked or high-risk items.
- Use closeout and signoffs to confirm review status, exports, and next actions.

Closeout item	Minimum operator action
Open tickets	Confirm status and next action.
Open positions	Review target/stop/max-hold plan.
Approvals/execution queue	Confirm no orphaned approvals or stale queued items.
Handoffs	Record unresolved context.
Escalations	Confirm severity and owner.
Live-readiness artifacts	Confirm no unintended live gates are enabled.
Exports/backups	Export critical CSVs or create backup.

11. Market Data Intelligence and Execution-Quality Simulation

Market data snapshots store local or optionally fetched order-book metrics. Public fetch is disabled by default. Execution-quality simulations are deterministic estimates, not fill guarantees.

Metric	Meaning	Operator use
best bid/ask	Top of book prices	Check spread and executable reference.
midpoint	Average of bid and ask	Approximate reference price.
spread bps	Bid/ask spread	Wide spreads increase execution risk.
top depth	Size near top of book	Thin depth can block/warn.
1%/5% depth	Depth within price bands	Liquidity assessment.
freshness	Snapshot age/status	Stale data should block live-readiness when required.

```
python -m app.cli --market-data-snapshots --json
python -m app.cli --record-market-data-snapshot --orderbook-json @orderbook.json --source manual --json
python -m app.cli --preview-execution-quality --market-id <market> --token-id <token> --side BUY --price 0.42 --size 10 --json
```

12. Live Readiness Architecture

Live readiness is staged so the operator can inspect every boundary before any real adapter call is possible. Each record reduces ambiguity and improves auditability.

45. Live config readiness.
46. Live order intent.
47. Live intent preflight.
48. Operator authorization.
49. Unsigned execution packet.
50. Dry-run adapter receipt.
51. Dry-run review.
52. Adapter request.
53. Manual execution review.
54. CLOB adapter status.
55. Manual submit/cancel record path.

Record or board	What it proves	What it does not prove
Live config readiness	Local environment values are present/missing/redacted.	Credentials valid or funded.
Live order intent	Operator drafted an order shape.	It is an order.
Live preflight	Intent passed/failed local gates.	Exchange validation.
Authorization	Human decision recorded.	Bypass stale data/risk gates.
Execution packet	Deterministic unsigned package exists.	Signed/submitted order.
Dry-run receipt	Local adapter-shaped validation passed/failed.	CLOB acknowledgement.
Adapter request	Request shape exists and gates were checked.	Submit is enabled.
Manual attempt	Preview/blocked/fake/real attempt record.	Only real adapter responses are exchange outcomes.

- `app/live_clob_adapter.py` is the only intended module for Polymarket trading SDK calls.
- Status screens and normal validation probe imports/methods without signing or network calls.
- Real submit/cancel use `py-clob-client` mapping behind gates.
- Install `requirements-live-optional.txt` only in live operator runtime.
- Do not duplicate SDK calls into routes, CLI, templates, or autonomous modules.

13. Manual Live Submit and Cancel SOPs

Follow this section exactly if evaluating live capability. Stop at the first failed gate.

13.1 Pre-live preparation checklist

- Review current Polymarket CLOB SDK/API documentation outside the app.
- Install optional live dependency only in the runtime.

- Confirm wallet, L2 credentials, token allowances, balances, and chain/funder/signature settings externally.
- Set small positive risk limits.
- Set market and token allowlists to exact intended ids.
- Set a unique final confirmation phrase.
- Keep kill switch active until the live window.
- Use fake adapter and dry-run workflows first.
- Export status/ledger CSVs before and after.

13.2 Required env gate pattern for real submit

```
POLYMARKET_LIVE_MODE=true
POLYMARKET_LIVE_ALLOW_REAL_NETWORK=true
POLYMARKET_LIVE_ENABLE_SUBMIT=true
POLYMARKET_LIVE_MANUAL_SUBMIT_ENABLED=true
POLYMARKET_LIVE_KILL_SWITCH=false
POLYMARKET_LIVE_MAX_ORDER_NOTIONAL=<small positive amount>
POLYMARKET_LIVE_MAX_DAILY_NOTIONAL=<small positive amount>
POLYMARKET_LIVE_MAX_OPEN_ORDERS=<small integer>
POLYMARKET_LIVE_MARKET_ALLOWLIST=<exact_market_id>
POLYMARKET_LIVE_TOKEN_ALLOWLIST=<exact_token_id>
POLYMARKET_LIVE_FINAL_CONFIRMATION_PHRASE=<unique local phrase>
```

13.3 Manual live submit SOP

56. Confirm blocked/default-off posture.
57. Verify upstream research, paper ticket, approval, preflight, market-data snapshot, and execution-quality simulation.
58. Create live order intent.
59. Run live intent preflight.
60. Record operator authorization.
61. Create unsigned execution packet.
62. Record dry-run receipt.
63. Create adapter request and confirm fields.
64. Open /live-clob-adapter and verify dependency/credential/method status.
65. Preview with adapter_mode blocked or fake_local first.
66. Set live env gates deliberately and restart if necessary.
67. Record with adapter_mode real_live and final confirmation phrase.
68. Inspect live order ledger and external Polymarket account/order view.
69. Re-enable kill switch after the live window.

13.4 Manual live cancel SOP

70. Confirm exchange order id externally and in local ledger if available.
71. Confirm cancellation policy.
72. Keep submit disabled unless separately planned.
73. Set cancel-specific gates only for intended cancel window.
74. Run cancel preview with order id, operator, reason, and confirmation.
75. Record cancel with adapter_mode real_live only if preview passes.
76. Inspect ledger and external exchange status.
77. Record reconciliation and incident/action notes.
78. Re-enable kill switch and disable cancel when done.

Emergency cancel mode is not a general bypass. It may allow cancellation while submit remains disabled, but must still require credentials, confirmation, audit, and real-network gates.

14. Live Order Ledger, Reconciliation, and Audit

The live order ledger records blocked attempts, fake-local events, and real adapter results when they occur. It stores hashes and redacted summaries instead of raw secrets.

Field family	Examples	Operator use
Identity	order_event_id, created_at, operator	Trace who did what and when.
Source links	intent_id, packet_id, adapter_request_id, execution_attempt_id	Trace an event back to source records.
Order shape	market_id, condition_id, token_id, side, price, size	Verify shape against intent.
Exchange fields	client_order_id, exchange_order_id, exchange_status	Compare local and remote state.
Safety flags	network_attempted, signed_payload_present, acknowledgement_present	Distinguish local/fake/real.
Integrity	raw_response_hash, event_hash	Tamper-evident review material.

Reconciliation state	Meaning	Action
reconciled	Local and remote views agree.	Archive/export.
local_only	Local record has no remote match.	Expected for blocked/fake; investigate if real attempted.
remote_only	Remote order not in local ledger.	Stop and investigate external/manual activity.
status_mismatch	Local status differs from remote.	Do not place duplicates; inspect exchange.
reconciliation_unavailable	Credentials/dependency/network unavailable.	Use local ledger and external exchange view.

15. Strategy Signals and Autonomous Bridge

Autonomous workflows remain intentionally constrained. The app can record deterministic strategy signals and preview/run dry-run or fake-adapter flows. It must not invent trades from vague LLM output.

Mode	Default status	Meaning
off	Default	No autonomous action.
dry_run	Allowed for previews	Evaluate deterministic signals without orders/network.
paper_only	Local simulation only	Research workflow, not live trading.
fake_adapter	Local simulation only	May route through fake adapter if enabled.
live_guarded	Blocked	Should remain blocked unless future release completes all autonomous gates.

- Strategy allowlist configured.
- Market/token allowlists configured.
- Max order notional and daily notional configured.
- Max orders per run/day configured.
- Signal not expired.
- Confidence threshold met.
- Market data and execution quality present when required.
- No severe unresolved alerts.
- No background scheduler unless explicitly enabled in a future reviewed release.

16. Daily and Weekly Operating Checklists

Checklist	Steps
Session start	Start server; log in; open /administration/config and /health; confirm live disabled, real network disabled, submit/cancel disabled, kill switch active; review /operator, /workflow, open positions, escalations, and handoffs.
Pre-trade paper	Review market; collect evidence/notes; evaluate playbook/readiness; create ticket; run preflight/risk; approve/reject; paper execute only when not blocked; document plan.
Pre-live	Review current SDK docs; verify wallet/balance/allowances externally; set tiny limits; exact allowlists; fresh market data/execution quality; upstream live records; fake-local workflow; export status; disable kill switch only during live window.
Closeout	Re-enable kill switch; disable submit/cancel; export audit/trades/positions/live orders/reconciliation/closeout CSVs; record handoff/signoff; create backup; confirm no secrets/state in source folder.

17. Backup, Maintenance, Upgrades, and Packaging Discipline

Runtime state lives under data/. It includes user database, session secret, paper ledgers, live ledgers, generated market data, generated simulations, and audit records. Treat data/ as sensitive.

- Use the administration maintenance page or filesystem backups.
- Encrypt backups that include data/.
- Store release ZIPs separately from runtime data.
- Before upgrading, stop the server and back up data/.
- After upgrading, run health, critical UI routes, and CLI smoke commands.

Never package	Why
.env	May contain private keys/API credentials.
data/users.json	Contains local user records.
data/session_secret.txt	Can invalidate or expose sessions.
data/live/* ledgers	Execution-adjacent operational history.
generated receipts/snapshots/simulations	Generated state.
.venv or venv	Large, non-portable, cache-heavy.
__pycache__, .pytest_cache	Generated cache; not source.

18. Troubleshooting and Incident Response

79. Record exact URL, time, app version, and operator action.
80. Check terminal logs for Python exception.
81. Do not continue live-adjacent operations until resolved.
82. Try matching API endpoint to isolate data vs template rendering.
83. Check malformed JSON under data/ if local state is involved.
84. Back up data/ before manual edits.
85. Validate route returns 200 and run compile/import checks after patching.

v1.0.1-real specifically hardened /positions and related APIs against malformed local paper portfolio data. If this page fails again, inspect data/paper files for non-numeric stake/shares/price fields and preserve the corrupt example for regression testing.

Likely live-submit blocker	Remediation
Kill switch active	Leave active unless in deliberate live window.
Submit disabled	Set submit flags only for deliberate window.
Real network disabled	Enable only with credentials, allowlists, limits, and confirmation.
Missing credentials	Populate only local .env.
Dependency missing	Install requirements-live-optional.txt in live runtime only.
Stale source records	Regenerate upstream records.
Market/token not allowlisted	Add exact ids only.
Confirmation mismatch	Use exact configured phrase.

19. CLI Reference

```
python -m app.cli --live-trading-status --json
python -m app.cli --live-clob-adapter-status --json
python -m app.cli --preview-live-submit --adapter-request-id <id> --adapter-mode blocked --operator <name> --json
python -m app.cli --preview-live-cancel --order-id <exchange_or_fake_order_id> --adapter-mode blocked --operator <name> --json
python -m app.cli --market-data-snapshots --json
python -m app.cli --execution-quality --json
python -m app.cli --paper-positions --json
python -m app.cli --audit-log --json
```

Workflow	Available flags
Research and market intelligence	--markets, --opportunities, --opportunity-engine, --search, --research, --watchlist, --add-watch, --remove-watch, --sources, --source-status, --evidence, --collect-evidence, --notes, --market-notes, --add-note, --delete-note

Workflow	Available flags
Paper workflow	--trade-tickets, --create-ticket, --ticket-detail, --update-ticket, --preflight, --approvals, --approve-ticket, --reject-ticket, --execution-queue, --paper-buy, --paper-sell, --portfolio, --trades, --paper-analytics

Workflow	Available flags
Risk/positions/settlement	--risk, --risk-check, --risk-budget, --paper-positions, --position-alerts, --set-position-plan, --exit-tickets, --create-exit-ticket, --execute-exit-ticket, --settlements, --settlement-candidates, --settle-market

Workflow	Available flags
Paper ops	--paper-ops-briefing, --briefing-checkpoints, --paper-handoffs, --handoff-reconciliation, --paper-ops-aging, --paper-ops-escalations, --paper-ops-escalation-review, --paper-ops-closeout, --paper-ops-closeout-signoffs, --runbook

Workflow	Available flags
Market data and execution quality	--market-data-snapshots, --record-market-data-snapshot, --

Workflow	Available flags
	parse-market-data-snapshot-preview, --fetch-market-data-snapshot-preview, --execution-quality, --preview-execution-quality, --record-execution-quality

Workflow	Available flags
Live readiness	--live-config-readiness, --live-order-intents, --preview-live-order-intent, --record-live-order-intent, --live-order-intent-preflight, --live-order-authorizations, --record-live-order-authorization, --live-execution-packets, --preview-live-execution-packet, --record-live-execution-packet, --live-dry-run-adapter, --record-live-dry-run-adapter, --live-adapter-requests, --record-live-adapter-request

Workflow	Available flags
Manual live boundary	--live-clob-adapter-status, --live-trading-status, --live-orders, --live-order-detail, --preview-live-submit, --record-live-submit, --preview-live-cancel, --record-live-cancel, --live-reconciliation, --record-live-reconciliation

Workflow	Available flags
Autonomous bridge	--strategy-signals, --record-strategy-signal, --validate-strategy-signal, --autonomous-trading-status, --preview-autonomous-run, --record-autonomous-run, --autonomous-runs

Workflow	Available flags
Shared/export	--json, --limit, --market-id, --condition-id, --token-id, --side, --price, --size, --operator, --adapter-mode, --final-confirmation, --export-live-orders, --export-live-reconciliation, --export-strategy-signals, --export-autonomous-runs

20. API and Route Reference

This reference focuses on operator-important routes. API routes return JSON or CSV/text unless otherwise stated. UI routes return HTML. Many endpoints require authentication depending on session state.

Operator console and workflow

Method	Route	Response
GET	/workflow	HTML
GET	/ui-system	HTML
GET	/	HTML
GET	/operator	HTML

Authentication and users

Method	Route	Response
GET	/setup	HTML
POST	/setup	JSON/form
GET	/login	HTML
POST	/login	JSON/form
GET	/logout	JSON/form
GET	/users	HTML

Method	Route	Response
POST	/users/create	JSON/form
POST	/users/update/{username}	JSON/form
POST	/users/delete/{username}	JSON/form
GET	/api/auth/me	JSON/form
GET	/api/users	JSON/form
POST	/api/users	JSON/form
DELETE	/api/users/{username}	JSON/form

Administration and runtime status

Method	Route	Response
GET	/administration/maintenance	HTML
POST	/administration/maintenance/backups/create	JSON/form
POST	/administration/maintenance/backups/delete/{filename}	JSON/form
GET	/api/maintenance/status	JSON/form
POST	/api/maintenance/backups	JSON/form
GET	/api/maintenance/backups/{filename}/download	JSON/form
DELETE	/api/maintenance/backups/{filename}	JSON/form
GET	/administration/config	HTML
GET	/api/deployment/status	JSON/form
GET	/api/status	JSON/form
GET	/api/network/status	JSON/form

Research, evidence, and market intelligence

Method	Route	Response
GET	/markets/{market_id}	HTML
GET	/research/sources	HTML
GET	/research/evidence	HTML
GET	/api/evidence	JSON/form
GET	/api/evidence/{packet_id}	JSON/form
GET	/api/markets/{market_id}/evidence-probability	JSON/form
GET	/api/evidence-probabilities	JSON/form
GET	/api/evidence/{packet_id}/score	JSON/form
GET	/api/markets/{market_id}/evidence-score	JSON/form
DELETE	/api/evidence/{packet_id}	JSON/form
POST	/api/markets/{market_id}/evidence-packet	JSON/form
POST	/markets/{market_id}/evidence-packet	JSON/form
GET	/api/markets/{market_id}/evidence-packets	JSON/form
GET	/api/sources	JSON/form
GET	/api/source-links	JSON/form
GET	/api/markets/{market_id}/source-pack	JSON/form
GET	/api/sources/status	JSON/form
GET	/api/markets/{market_id}/collection-targets	JSON/form
GET	/api/markets/{market_id}/links	JSON/form
GET	/api/notes	JSON/form

Paper trading, risk, and operations

Method	Route	Response
GET	/api/paper/analytics	JSON/form
GET	/api/paper/trades.csv	CSV/text
GET	/api/paper/trades	JSON/form
GET	/api/paper/audit	JSON/form
GET	/api/paper/audit.csv	CSV/text
GET	/api/paper/audit/{market_id}	JSON/form
GET	/audit	HTML
GET	/api/paper/review-report	JSON/form
GET	/api/paper/review-report.csv	CSV/text
GET	/api/paper/review-report/{market_id}	JSON/form
GET	/review-report	HTML
GET	/api/paper/positions	JSON/form
GET	/api/paper/position-alerts	JSON/form
POST	/api/paper/positions/{market_id}/plan	JSON/form
GET	/positions	HTML
POST	/paper/positions/plan	JSON/form
POST	/api/paper/reset	JSON/form
POST	/api/paper/buy/{market_id}	JSON/form
POST	/api/paper/sell/{market_id}	JSON/form
GET	/api/paper/settlements	JSON/form

Playbooks, tickets, and exits

Method	Route	Response
POST	/api/trade-tickets/{ticket_id}/preflight	JSON/form
POST	/api/trade-tickets/{ticket_id}/approval	JSON/form
POST	/trade-tickets/{ticket_id}/approval	JSON/form
POST	/api/trade-tickets/{ticket_id}/reject	JSON/form
POST	/trade-tickets/{ticket_id}/reject	JSON/form
GET	/api/playbooks	JSON/form
GET	/api/playbooks/board	JSON/form
GET	/api/playbook-decisions	JSON/form
GET	/api/playbooks/{playbook_id}	JSON/form
POST	/playbooks/custom	JSON/form
GET	/playbooks	HTML
GET	/api/playbook-performance	JSON/form
GET	/api/playbook-performance.csv	CSV/text
GET	/api/playbook-performance/{playbook_id}	JSON/form
GET	/playbook-performance	HTML
GET	/api/trade-tickets	JSON/form
GET	/trade-tickets	HTML
GET	/api/trade-tickets/{ticket_id}	JSON/form
POST	/api/trade-tickets/{ticket_id}/status	JSON/form
POST	/trade-tickets/{ticket_id}/status	JSON/form

Market data and execution quality

Method	Route	Response
GET	/api/market-data/snapshots	JSON/form
GET	/api/market-data/snapshots.csv	CSV/text
GET	/api/market-data/snapshots/{snapshot_id}	JSON/form
POST	/api/market-data/snapshots/parse-preview	JSON/form
POST	/api/market-data/snapshots	JSON/form
POST	/api/market-data/snapshots/fetch-	JSON/form

Method	Route	Response
	preview	
POST	/api/market-data/snapshots/fetch-record	JSON/form
GET	/api/execution-quality	JSON/form
GET	/api/execution-quality.csv	CSV/text
GET	/api/execution-quality/{simulation_id}	JSON/form
POST	/api/execution-quality/preview	JSON/form
POST	/api/execution-quality/record	JSON/form
GET	/market-data	HTML
GET	/market-data/snapshots	HTML
GET	/market-data/snapshots/{snapshot_id}	HTML
GET	/execution-quality	HTML
GET	/execution-quality/{simulation_id}	HTML

Live readiness and execution control

Method	Route	Response
GET	/api/live/config/readiness	JSON/form
GET	/api/live/config/readiness.csv	JSON/form
GET	/api/live/config/template.env	JSON/form
GET	/api/live/dry-run-adapter	JSON/form
GET	/api/live/dry-run-adapter.csv	CSV/text
GET	/api/live/dry-run-adapter/{receipt_id}	JSON/form
GET	/api/live/dry-run-review	JSON/form
GET	/api/live/dry-run-review.csv	CSV/text
GET	/api/live/dry-run-review/{packet_id}	JSON/form
POST	/api/live/execution-packets/{packet_id}/dry-run/preview	JSON/form
POST	/api/live/execution-packets/{packet_id}/dry-run	JSON/form
GET	/api/live/adapter/readiness	JSON/form
GET	/api/live/adapter/readiness.csv	CSV/text
GET	/api/live/adapter/readonly-validations	JSON/form
GET	/api/live/adapter/readonly-validations.csv	CSV/text
GET	/api/live/adapter/readonly-validations/{validation_id}	JSON/form
POST	/api/live/adapter/readonly-validation/preview	JSON/form
POST	/api/live/adapter/readonly-validation	JSON/form
GET	/api/live/adapter/requests	JSON/form
GET	/api/live/adapter/requests.csv	CSV/text
GET	/api/live/adapter/requests/{request_or_packet_id}	JSON/form
POST	/api/live/execution-packets/{packet_id}/adapter-request/preview	JSON/form
POST	/api/live/execution-packets/{packet_id}/adapter-request	JSON/form
GET	/api/live/manual-execution-reviews	JSON/form
GET	/api/live/manual-execution-reviews.csv	CSV/text
GET	/api/live/manual-execution-reviews/{review_id}	JSON/form
POST	/api/live/execution-packets/{packet_id}/manual-execution-review/preview	JSON/form
POST	/api/live/execution-packets/{packet_id}/manual-execution-review	JSON/form
GET	/api/live/execution-control/readiness	JSON/form
GET	/api/live/execution-control/readiness.csv	CSV/text
GET	/api/live/execution-attempts	JSON/form
GET	/api/live/execution-attempts.csv	CSV/text
GET	/api/live/execution-attempts/{attempt_id}	JSON/form

Method	Route	Response
POST	/api/live/adapter/requests/{adapter_request_id}/manual-submit/preview	JSON/form
POST	/api/live/adapter/requests/{adapter_request_id}/manual-submit	JSON/form

Strategy and autonomous bridge

Method	Route	Response
GET	/strategy-signals	HTML
GET	/autonomous-trading	HTML
GET	/autonomous-runs	HTML
GET	/api/strategy-signals	JSON/form
GET	/api/strategy-signals.csv	CSV/text
GET	/api/strategy-signals/{signal_id}	JSON/form
POST	/api/strategy-signals/validate-preview	JSON/form
POST	/api/strategy-signals	JSON/form
GET	/api/autonomous-trading/status	JSON/form
POST	/api/autonomous-trading/run-preview	JSON/form
POST	/api/autonomous-trading/run	JSON/form
GET	/api/autonomous-runs	JSON/form
GET	/api/autonomous-runs.csv	CSV/text
GET	/api/autonomous-runs/{run_id}	JSON/form
GET	/api/strategy	JSON/form
GET	/api/strategy/recommendations	JSON/form

20.1 Live route quick map

Method	Live route	Response
GET	/api/live/config/readiness	JSON/form
GET	/api/live/config/readiness.csv	JSON/form
GET	/api/live/config/template.env	JSON/form
GET	/api/live/dry-run-adapter	JSON/form
GET	/api/live/dry-run-adapter.csv	CSV/text
GET	/api/live/dry-run-adapter/{receipt_id}	JSON/form
GET	/api/live/dry-run-review	JSON/form
GET	/api/live/dry-run-review.csv	CSV/text
GET	/api/live/dry-run-review/{packet_id}	JSON/form
POST	/api/live/execution-packets/{packet_id}/dry-run/preview	JSON/form
POST	/api/live/execution-packets/{packet_id}/dry-run	JSON/form
GET	/api/live/adapter/readiness	JSON/form
GET	/api/live/adapter/readiness.csv	CSV/text
GET	/api/live/adapter/readonly-validations	JSON/form
GET	/api/live/adapter/readonly-validations.csv	CSV/text
GET	/api/live/adapter/readonly-validations/{validation_id}	JSON/form
POST	/api/live/adapter/readonly-validation/preview	JSON/form
POST	/api/live/adapter/readonly-validation	JSON/form
GET	/api/live/adapter/requests	JSON/form
GET	/api/live/adapter/requests.csv	CSV/text
GET	/api/live/adapter/requests/{request_or_packet_id}	JSON/form
POST	/api/live/execution-packets/{packet_id}/adapter-request/preview	JSON/form
POST	/api/live/execution-packets/{packet_id}/	JSON/form

Method	Live route	Response
	adapter-request	
GET	/api/live/manual-execution-reviews	JSON/form
GET	/api/live/manual-execution-reviews.csv	CSV/text
GET	/api/live/manual-execution-reviews/{review_id}	JSON/form
POST	/api/live/execution-packets/{packet_id}/manual-execution-review/preview	JSON/form
POST	/api/live/execution-packets/{packet_id}/manual-execution-review	JSON/form
GET	/api/live/execution-control/readiness	JSON/form
GET	/api/live/execution-control/readiness.csv	CSV/text
GET	/api/live/execution-attempts	JSON/form
GET	/api/live/execution-attempts.csv	CSV/text
GET	/api/live/execution-attempts/{attempt_id}	JSON/form
POST	/api/live/adapter/requests/{adapter_request_id}/manual-submit/preview	JSON/form
POST	/api/live/adapter/requests/{adapter_request_id}/manual-submit	JSON/form
POST	/api/live/manual-cancel/preview	JSON/form
POST	/api/live/manual-cancel	JSON/form
GET	/live-clob-adapter	HTML
GET	/api/live/clob-adapter/status	JSON/form
GET	/api/live/clob-adapter/status.csv	CSV/text
GET	/live-trading	HTML
GET	/live-orders	HTML
GET	/live-reconciliation	HTML
GET	/api/live/trading/status	JSON/form
GET	/api/live/orders	JSON/form
GET	/api/live/orders.csv	CSV/text
GET	/api/live/orders/{order_event_id}	JSON/form
POST	/api/live/adapter/requests/{adapter_request_id}/manual-submit-preview	JSON/form
POST	/api/live/orders/{order_id}/cancel-preview	JSON/form
POST	/api/live/orders/{order_id}/cancel	JSON/form
POST	/api/live/reconciliation/preview	JSON/form
POST	/api/live/reconciliation/record	JSON/form
GET	/api/live/reconciliation	JSON/form
GET	/api/live/reconciliation.csv	CSV/text
GET	/live-config	HTML
GET	/api/live/order-intents	JSON/form
GET	/api/live/order-intents.csv	CSV/text
GET	/api/live/order-intents/preflight	JSON/form
GET	/api/live/order-intents/preflight.csv	CSV/text
GET	/api/live/order-intents/{intent_id}/preflight	JSON/form
GET	/api/live/order-intents/authorizations	JSON/form
GET	/api/live/order-intents/authorizations.csv	CSV/text
GET	/api/live/order-intents/authorizations/{authorization_id}	JSON/form
POST	/api/live/order-intents/{intent_id}/authorization	JSON/form
GET	/api/live/execution-packets	JSON/form
GET	/api/live/execution-packets.csv	CSV/text
GET	/api/live/execution-packets/{packet_id}	JSON/form
POST	/api/live/order-intents/{intent_id}/execution-packet	JSON/form

Method	Live route	Response
POST	/api/live/order-intents/{intent_id}/ execution-packet/preview	JSON/form
GET	/api/live/order-intents/{intent_id}	JSON/form

21. Data Files and Secret-Handling Appendix

Path pattern	Contents	Handling
data/users.json	Local users and password hashes.	Sensitive; backup encrypted; exclude from release ZIP.
data/session_secret.txt	Session signing secret.	Sensitive; exclude from release ZIP.
data/paper/*	Paper trades, portfolio, tickets, approvals, ops records.	Operational state; backup; exclude from release ZIP.
data/live/*	Live intents, packets, receipts, adapter requests, attempts, order events, signals.	Execution-adjacent state; backup securely; exclude from release ZIP.
data/snapshots/*	Market snapshots and fetched/manual records.	Generated state; exclude from release ZIP.
.env	Local runtime secrets and settings.	Never share or package.
requirements-live-optional.txt	Optional live SDK dependency list.	Safe to package; review before install.

- Status pages should show presence booleans or masked strings only.
- Errors should be redacted before being returned in JSON/CSV/HTML/log summaries.
- Response hashes are acceptable; raw secrets/private keys/full signed payloads should not be stored by default.
- When sending logs for help, inspect them for credentials first.

22. Glossary

Term	Definition
Adapter boundary	The isolated module layer through which real CLOB SDK calls must pass.
Adapter request	A local adapter-shaped request record derived from an execution packet.
Audit ledger	Local event history used for review and evidence.
Autonomous bridge	Dry-run/fake-adapter deterministic signal workflow; not live autonomous trading by default.
CLOB	Central limit order book. In this project, refers to Polymarket order interaction surfaces.
Dry-run receipt	Local no-network adapter simulation receipt. Not an exchange acknowledgement.
Execution packet	Unsigned local package of intent/preflight/authorization fields. Not an order.
Fake adapter	Local simulator for submit/cancel shaped events. Not live trading.
Final confirmation phrase	Configured phrase that must match operator input before live record actions can proceed.
Kill switch	Hard safety gate that blocks live submit/cancel by default.
Live order intent	Local record of intended live order shape. Not an order.
Live order ledger	Local ledger of live-adjacent order events, blocked/fake/real.
Market allowlist	Explicit list of markets permitted for live/autonomous gates.
Preflight	Local check that validates order, risk, and state conditions before proceeding.
Reconciliation	Read-only comparison between local ledger and available remote/order status.
Strategy signal	Explicit deterministic local record consumed by autonomous dry-run/fake-adapter logic.

Term	Definition
Token allowlist	Explicit list of Polymarket token ids permitted for live gates.

Appendix A: Selected Environment Variables

Key	Default/example	Section/comment
HOST	0.0.0.0	Use HOST=127.0.0.1 to make it local-only again.
PORT	8000	Use HOST=127.0.0.1 to make it local-only again.
APP_RELOAD	true	Use HOST=127.0.0.1 to make it local-only again.
ALLOWED_HOSTS	*	LAN/security controls. Use * for easy LAN testing, or set fixed hosts/IPs.
SECURITY_HEADERS_ENABLED	true	LAN/security controls. Use * for easy LAN testing, or set fixed hosts/IPs.
SESSION_COOKIE_SECURE	false	LAN/security controls. Use * for easy LAN testing, or set fixed hosts/IPs.
SESSION_COOKIE_SAMESITE	lax	LAN/security controls. Use * for easy LAN testing, or set fixed hosts/IPs.
GAMMA_BASE_URL	https://gamma-api.polymarket.com	LAN/security controls. Use * for easy LAN testing, or set fixed hosts/IPs.
CLOB_BASE_URL	https://clob.polymarket.com	LAN/security controls. Use * for easy LAN testing, or set fixed hosts/IPs.
REQUEST_TIMEOUT_SECONDS	20	LAN/security controls. Use * for easy LAN testing, or set fixed hosts/IPs.
DEFAULT_LIMIT	20	LAN/security controls. Use * for easy LAN testing, or set fixed hosts/IPs.
APP_MODE	read_only	LAN/security controls. Use * for easy LAN testing, or set fixed hosts/IPs.
READ_ONLY	true	LAN/security controls. Use * for easy LAN testing, or set fixed hosts/IPs.
LIVE_TRADING_ENABLED	false	LAN/security controls. Use * for easy LAN testing, or set fixed hosts/IPs.
PAPER_MAX_STAKE_PER_TRADE	250	LAN/security controls. Use * for easy LAN testing, or set fixed hosts/IPs.
PAPER_MAX_MARKET_EXPOSURE	500	LAN/security controls. Use * for easy LAN testing, or set fixed hosts/IPs.
PAPER_MAX_TOTAL_EXPOSURE	2500	LAN/security controls. Use * for easy LAN testing, or set fixed hosts/IPs.
PAPER_MAX_OPEN_POSITIONS	20	LAN/security controls. Use * for easy LAN testing, or set fixed hosts/IPs.
PAPER_MIN_LIQUIDITY	1000	LAN/security controls. Use * for easy LAN testing, or set fixed hosts/IPs.
PAPER_MIN_VOLUME_24HR	10	LAN/security controls. Use * for easy LAN testing, or set fixed hosts/IPs.
PAPER_BLOCK_EXTREME_PRICES	true	LAN/security controls. Use * for easy LAN testing, or set fixed hosts/IPs.
PAPER_MIN_PRICE	0.02	LAN/security controls. Use * for easy LAN testing, or set fixed hosts/IPs.
PAPER_MAX_PRICE	0.98	LAN/security controls. Use * for easy LAN testing, or set fixed hosts/IPs.
OPENAI_API_KEY		Future optional integrations. Not needed for the current local paper workflow.
NEWS_API_KEY		Future optional integrations. Not needed for the current local paper workflow.
LIVE_DRY_RUN_ONLY	true	or automate live trading.
LIVE_REQUIRE_MANUAL_APPROVAL	true	or automate live trading.
LIVE_PRETRADE_CHECKS_ENABLED	true	or automate live trading.

Key	Default/example	Section/comment
LIVE_AUDIT_REQUIRED	true	or automate live trading.
LIVE_MAX_ORDER_NOTIONAL	0	or automate live trading.
LIVE_MAX_MARKET_NOTIONAL	0	or automate live trading.
LIVE_MAX_DAILY_NOTIONAL	0	or automate live trading.
LIVE_MAX_OPEN_ORDERS	0	or automate live trading.
LIVE_ALLOWED_MARKET_IDS		or automate live trading.
POLYMARKET_LIVE_MODE	false	Live adapter boundary. All execution-facing controls are default-off.
POLYMARKET_LIVE_NETWORK_READONLY	false	Live adapter boundary. All execution-facing controls are default-off.
POLYMARKET_LIVE_ENABLE_SUBMIT	false	Live adapter boundary. All execution-facing controls are default-off.
POLYMARKET_LIVE_ENABLE_CANCEL	false	Live adapter boundary. All execution-facing controls are default-off.
POLYMARKET_LIVE_REQUIRE_MANUAL_AUTH	true	Live adapter boundary. All execution-facing controls are default-off.
POLYMARKET_LIVE_KILL_SWITCH	true	Live adapter boundary. All execution-facing controls are default-off.
POLYMARKET_LIVE_REQUIRE_DRY_RUN_RECEIPT	true	Live adapter boundary. All execution-facing controls are default-off.
POLYMARKET_LIVE_READONLY_TIMEOUT_SECONDS	4	Live adapter boundary. All execution-facing controls are default-off.
POLYMARKET_CLOB_HOST	https://clob.polymarket.com	Live adapter boundary. All execution-facing controls are default-off.
POLYMARKET_LIVE_MANUAL_SUBMIT_ENABLED	false	Fake receipts are local simulations only. Real live submit/cancel are implemented
POLYMARKET_LIVE_MANUAL_CANCEL_ENABLED	false	Fake receipts are local simulations only. Real live submit/cancel are implemented
POLYMARKET_LIVE_FAKE_ADAPTER_ENABLED	false	Fake receipts are local simulations only. Real live submit/cancel are implemented
POLYMARKET_LIVE_REAL_ADAPTER_EXPERIMENTAL	false	Fake receipts are local simulations only. Real live submit/cancel are implemented
POLYMARKET_RUN_REAL_LIVE_TESTS	false	Fake receipts are local simulations only. Real live submit/cancel are implemented
POLYMARKET_REAL_LIVE_TEST_CONFIRMATION		Fake receipts are local simulations only. Real live submit/cancel are implemented
POLYMARKET_LIVE_FINAL_CONFIRMATION_PHRASE		Fake receipts are local simulations only. Real live submit/cancel are implemented
POLYMARKET_LIVE_AUTHORIZATION_MAX_AGE_MINUTES	60	Fake receipts are local simulations only. Real live submit/cancel are implemented
POLYMARKET_LIVE_DRY_RUN_MAX_AGE_MINUTES	60	Fake receipts are local simulations only. Real live submit/cancel are implemented
POLYMARKET_LIVE_ADAPTER_REQUEST_MAX_AGE_MINUTES	60	Fake receipts are local simulations only. Real live submit/cancel are implemented
POLYMARKET_MARKET_DATA_REQUIRE_FOR_LIVE	true	Public fetch remains disabled by default; local/manual JSON fixtures are support
POLYMARKET_MARKET_DATA_MAX_AGE_SECONDS	300	Public fetch remains disabled by default; local/manual JSON fixtures are support
POLYMARKET_MARKET_DATA_MAX_SPREAD_BPS	250	Public fetch remains disabled by default; local/manual JSON fixtures are support
POLYMARKET_MARKET_DATA_MAX_SLIPPAGE_BPS	150	Public fetch remains disabled by default; local/manual JSON fixtures are support
POLYMARKET_MARKET_DATA_MIN_TOP_DEPTH	10	Public fetch remains disabled by default; local/manual JSON fixtures are support
POLYMARKET_MARKET_DATA_MIN_TOTAL_DEPTH	50	Public fetch remains disabled by default; local/manual JSON fixtures are support
POLYMARKET_MARKET_DATA_PUBLIC_FETCH_ENABLED	false	Public fetch remains disabled by default; local/manual JSON fixtures are support
POLYMARKET_MARKET_DATA_TIMEOUT_SECONDS	4	Public fetch remains disabled by default; local/manual JSON fixtures are support

Key	Default/example	Section/comment
POLYMARKET_LIVE_ALLOW_REAL_NETWORK	false	Real submit/cancel remain fail-closed until a future release maps and validates
POLYMARKET_CHAIN_ID	137	Populate only in your local .env, never in a shared ZIP.
POLYMARKET_SIGNATURE_TYPE		Populate only in your local .env, never in a shared ZIP.
POLYMARKET_FUNDER_ADDRESS		Populate only in your local .env, never in a shared ZIP.
POLY_ADDRESS		Populate only in your local .env, never in a shared ZIP.
POLYMARKET_WALLET_ADDRESS		Populate only in your local .env, never in a shared ZIP.
POLY_PRIVATE_KEY		Populate only in your local .env, never in a shared ZIP.
POLYMARKET_PRIVATE_KEY		Populate only in your local .env, never in a shared ZIP.
POLY_API_KEY		Populate only in your local .env, never in a shared ZIP.
POLYMARKET_CLOB_API_KEY		Populate only in your local .env, never in a shared ZIP.
CLOB_API_KEY		Populate only in your local .env, never in a shared ZIP.
POLY_SECRET		Populate only in your local .env, never in a shared ZIP.
POLYMARKET_CLOB_SECRET		Populate only in your local .env, never in a shared ZIP.
CLOB_SECRET		Populate only in your local .env, never in a shared ZIP.
POLY_PASSPHRASE		Populate only in your local .env, never in a shared ZIP.
POLYMARKET_CLOB_PASSPHRASE		Populate only in your local .env, never in a shared ZIP.
CLOB_PASSPHRASE		Populate only in your local .env, never in a shared ZIP.
SOURCE_CHECK_TIMEOUT	4	Research source availability checks
POLYMARKET_LIVE_ENABLE_AUTONOMOUS	false	v1.0.0 guarded live/autonomous bridge controls (safe defaults; duplicated names)
POLYMARKET_LIVE_MAX_ORDER_NOTIONAL	0	v1.0.0 guarded live/autonomous bridge controls (safe defaults; duplicated names)
POLYMARKET_LIVE_MAX_DAILY_NOTIONAL	0	v1.0.0 guarded live/autonomous bridge controls (safe defaults; duplicated names)
POLYMARKET_LIVE_MAX_OPEN_ORDERS	0	v1.0.0 guarded live/autonomous bridge controls (safe defaults; duplicated names)
POLYMARKET_LIVE_MAX_POSITION_NOTIONAL	0	v1.0.0 guarded live/autonomous bridge controls (safe defaults; duplicated names)
POLYMARKET_LIVE_MAX_DAILY_LOSS	0	v1.0.0 guarded live/autonomous bridge controls (safe defaults; duplicated names)
POLYMARKET_LIVE_MARKET_ALLOWLIST		v1.0.0 guarded live/autonomous bridge controls (safe defaults; duplicated names)
POLYMARKET_LIVE_TOKEN_ALLOWLIST		v1.0.0 guarded live/autonomous bridge controls (safe defaults; duplicated names)
POLYMARKET_LIVE_EMERGENCY_CANCEL_ENABLED	false	v1.0.0 guarded live/autonomous bridge controls (safe defaults; duplicated names)
POLYMARKET_AUTONOMOUS_MAX_ORDERS_PER_RUN	0	v1.0.0 guarded live/autonomous bridge controls (safe defaults; duplicated names)
POLYMARKET_AUTONOMOUS_MAX_ORDERS_PER_DAY	0	v1.0.0 guarded live/autonomous bridge controls (safe defaults; duplicated names)
POLYMARKET_AUTONOMOUS_MIN_SIGNAL_CONFIDENCE		v1.0.0 guarded live/autonomous bridge controls (safe defaults; duplicated names)
POLYMARKET_AUTONOMOUS_REQUIRE_	true	v1.0.0 guarded live/autonomous bridge

Key	Default/example	Section/comment
MARKET_DATA		controls (safe defaults; duplicated names)
POLYMARKET_AUTONOMOUS_REQUIRE_EXECUTION_QUALITY	true	v1.0.0 guarded live/autonomous bridge controls (safe defaults; duplicated names)
POLYMARKET_AUTONOMOUS_REQUIRE_PAPER_APPROVAL	true	v1.0.0 guarded live/autonomous bridge controls (safe defaults; duplicated names)
POLYMARKET_AUTONOMOUS_STRATEGY_ALLOWLIST		v1.0.0 guarded live/autonomous bridge controls (safe defaults; duplicated names)
POLYMARKET_AUTONOMOUS_DRY_RUN_BY_DEFAULT	true	v1.0.0 guarded live/autonomous bridge controls (safe defaults; duplicated names)
POLYMARKET_AUTONOMOUS_SCHEDULER_ENABLED	false	v1.0.0 guarded live/autonomous bridge controls (safe defaults; duplicated names)
POLYMARKET_AUTONOMOUS_SCHEDULER_INTERVAL_SECONDS	0	v1.0.0 guarded live/autonomous bridge controls (safe defaults; duplicated names)
POLYMARKET_AUTONOMOUS_SCHEDULER_DRY_RUN_ONLY	true	v1.0.0 guarded live/autonomous bridge controls (safe defaults; duplicated names)
POLYMARKET_LIVE_SDK_FAMILY	current_py_clob_client	v1.0.0 SDK mapping probe controls (safe defaults; no real values)
POLYMARKET_LIVE_SDK_SUBMIT_METHOD	create_order+post_order	v1.0.0 SDK mapping probe controls (safe defaults; no real values)
POLYMARKET_LIVE_SDK_CANCEL_METHOD	cancel	v1.0.0 SDK mapping probe controls (safe defaults; no real values)

Appendix B: Validation Commands

```
python -m compileall app
python - <<PY
from app.main import app
print(app.title, app.version)
PY
python -m app.cli --live-trading-status --json
python -m app.cli --live-clob-adapter-status --json
python -m app.cli --preview-live-submit --adapter-request-id synthetic --adapter-mode blocked --operator manual --json
python -m app.cli --preview-live-cancel --order-id synthetic --adapter-mode blocked --operator manual --json
```

Appendix C: Manual Live Enablement Decision Record Template

Field	Operator entry
Date/time UTC	
Operator	
Package version	1.0.1-real
Market id	
Token id	
Side/price/size	
Maximum notional	
Daily notional remaining	
External wallet/balance verified	
Current SDK docs reviewed	
Optional live dependency installed	
Market data snapshot id	
Execution-quality simulation id	
Intent/preflight/authorization/packet ids	
Dry-run receipt id	
Adapter request id	
Kill switch disabled at	

Field	Operator entry
Final confirmation phrase matched	
Live order ledger event id	
External order id/status	
Kill switch re-enabled at	
Post-action notes	

Final Operator Reminder

The safest successful operation is often a blocked operation that explains exactly why it is blocked. Do not treat blocked states as obstacles to bypass. Treat them as the control system doing its job.

- Keep live disabled until a deliberate live window.
- Keep the kill switch active unless every gate has been reviewed.
- Prefer paper, dry-run, and fake-adapter validation until live operation is necessary and justified.
- Never package secrets or generated state.
- Export audit records after every material session.
- When in doubt, stop and reconcile.